Embedded rain-sensing system

In a rain-sensing system, an optical sensor is placed on a small area on the front windshield glass opposite to the rear-view mirror. This optical sensor emits infrared light and is placed at a 45-degree angle to the windshield. If the glass is dry, most of this light is reflected back into the sensor. If water droplets are on the glass, these reflect light in different directions. The wetter the glass, the lesser the light that makes back to the sensor.

The electronics and software in the sensor turn on the wipers when the amount of light reflected onto the sensor decreases to a preset level. The software sets the speed of the wipers based on how fast the moisture builds up between wipes. It can operate the wipers at any speed. The system adjusts the speed as often as necessary to match with the rate of moisture accumulation.

Embedded based automatic car parking system

This automatic car parking system is an independent car-manipulation system that moves a car from a traffic lane into a parking spot to perform parallel, perpendicular and angle parking.

The system mainly uses different methods to detect objects around the car. Sensors installed on the front of the vehicle and rear bumpers act as both a transmitter and a receiver. These send a signal that is replicated back when it meets an obstacle near the vehicle and then the car computer receives the time signal and the bumper uses the radar to decide the position of the obstacle. The car senses the parking space and distance from the side of the road and helps the driver drive the car into the parking place.

Future prospects

With the ever-increasing use of embedded systems in automobiles, new technologies and enhancements are being developed to increase usability.

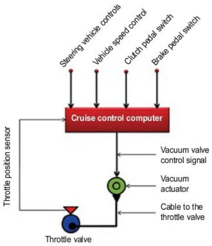

Fig. 12: Working of a cruise control system

Fig. 13: Drive by wire concept car

Intelligent adaptive cruise control. By using this technology we can make driverless vehicle control a reality. Many automobile manufacturers are already engaged with this concept. This adaptive cruise control allows cars to keep safe distances from other vehicles on busy highways. The driver of the car can set the speed of the vehicle and the distance between his car and other vehicles. When traffic slows down, adaptive cruise control changes vehicle speed using moderate braking.

Each car has a laser transceiver or a microwave radar unit, which is fixed in front of the car to find out the speed and distance of the any other vehicle in the pathway. This works on the principle of Doppler Effect, which is basically change in the frequency of waves. It uses a forward-facing radar, installed behind the grill of a vehicle, to detect the speed and the vehicle ahead of it. It can automatically adjust speed in order to maintain proper distance between vehicles in the same lane.

It is hard to implement this system around a curve, when there are elevation changes, during cut-ins, during dense traffic and other situations.

Drive by wire. Drive by wire system replaces mechanical connections like push rods, rack and pinion, steering columns, overhead cams and cables by mechatronic connections like sensors, actuators, embedded microprocessors and control software.

Throttle by wire. This system helps accomplish vehicle propulsion by means of an electric throttle without any cables from the accelerator pedal to the throttle valve at the engine. It controls electric motors by sensing accelerator pedal input and sending commands to power inverter modules.

Brake by wire. A pure brake by wire system eliminates the need for hydraulics completely by using motors to actuate calipers and lock the wheels in comparison to the current technology where the system is designed to provide braking effort by building hydraulic pressure in brake lines.

Shift by wire. Direction of motion of the vehicle (forward or reverse) is set by commanding the actuators inside the transmission through electronic commands based on current input from the driver (park, reverse, neutral or drive).

Steer by wire. This system provides steering control of a car with fewer mechanical components/linkages between the steering wheel and the wheels. Control of the wheels' direction is established through electric motor(s), which are actuated by electronic control units monitoring steering wheel inputs from the driver. EFY